

**Electronics and Communication Engineering (Dual Degree) - Semester wise Plan (Subject to minor changes)**

| Year & Sem | Course Code  | Course Name                                | Full Sem / Half Sem | Lectures(L), Tutorial(T), Practical(P)<br>Hours per week |   |    | Credits (C) |           |           |       |         |     |           |           |
|------------|--------------|--------------------------------------------|---------------------|----------------------------------------------------------|---|----|-------------|-----------|-----------|-------|---------|-----|-----------|-----------|
|            |              |                                            |                     | L                                                        | T | P  | Ins Core    | Prog Core | Prog Elec | Maths | Science | Hum | Open Elec | Hons      |
| I-I        | Monsoon      |                                            |                     |                                                          |   |    |             |           |           |       |         |     |           |           |
|            | EC5.101      | Networks, Signals and Systems              | Full                | 3                                                        | 1 | 0  | -           | 4         | -         | -     | -       | -   | -         | -         |
|            | MA4.101      | Real Analysis                              | Full                | 3                                                        | 1 | 0  | -           | -         | -         | 4     | -       | -   | -         | -         |
|            | EC2.101      | Digital Systems and Microcontrollers (DSM) | Full                | 3                                                        | 1 | 3  | -           | 5         | -         | -     | -       | -   | -         | -         |
|            | CS0.101      | Computer Programming                       | Full                | 3                                                        | 1 | 3  | -           | 5         | -         | -     | -       | -   | -         | -         |
|            | EC2.102      | Electronic Workshop-1                      | Half                | 0                                                        | 1 | 3  | -           | 2         | -         | -     | -       | -   | -         | -         |
|            | OC2.101      | Arts-1                                     | Half                | 0                                                        | 2 | 2  | 2           | -         | -         | -     | -       | -   | -         | -         |
|            | OC1.101      | Sports-1                                   | Full                | 0                                                        | 0 | 2  | 1           | -         | -         | -     | -       | -   | -         | -         |
|            | OC3.101      | Value Education-1                          | Full                | 2                                                        | 1 | 0  | 2           | -         | -         | -     | -       | -   | -         | -         |
|            |              | <b>Sub Total</b>                           |                     | 14                                                       | 8 | 13 | 5           | 16        | 0         | 4     | 0       | 0   | 0         | 0         |
|            | <b>Total</b> |                                            |                     |                                                          |   |    |             |           |           |       |         |     |           | <b>25</b> |
| I-II       | Spring       |                                            |                     |                                                          |   |    |             |           |           |       |         |     |           |           |
|            | CS1.201      | Data Structures and Algorithms             | Full                | 3                                                        | 1 | 3  | -           | 5         | -         | -     | -       | -   | -         | -         |
|            | MA2.101      | Linear Algebra                             | Full                | 3                                                        | 1 | 0  | -           | -         | -         | 4     | -       | -   | -         | -         |
|            | EC2.103      | Analog Electronic Circuits                 | Full                | 3                                                        | 1 | 3  | -           | 5         | -         | -     | -       | -   | -         | -         |
|            | EC5.102      | Information and Communication              | Full                | 3                                                        | 1 | 0  | -           | 4         | -         | -     | -       | -   | -         | -         |
|            | CS1.102      | Sports-2                                   | Full                | 0                                                        | 0 | 2  | 1           | -         | -         | -     | -       | -   | -         | -         |
|            | CS2.102      | Arts-2                                     | Full                | 0                                                        | 2 | 2  | 2           | -         | -         | -     | -       | -   | -         | -         |
|            |              | <b>Sub Total</b>                           |                     | 12                                                       | 6 | 10 | 3           | 14        | 0         | 4     | 0       | 0   | 0         | 0         |
|            | <b>Total</b> |                                            |                     |                                                          |   |    |             |           |           |       |         |     |           | <b>21</b> |
| II-I       | Monsoon      |                                            |                     |                                                          |   |    |             |           |           |       |         |     |           |           |
|            | MA6.102      | Probability and Random Processes           | Full                | 3                                                        | 1 | 0  | -           | -         | -         | 4     | -       | -   | -         | -         |
|            | EC5.201      | Signal Processing                          | Full                | 3                                                        | 1 | 3  | -           | 5         | -         | -     | -       | -   | -         | -         |
|            | EC5.202      | Systems Thinking (Commn, bio, neuro)       | Full                | 3                                                        | 1 | 0  | -           | 4         | -         | -     | -       | -   | -         | -         |
|            | EC6.201      | VLSI Design                                | Full                | 3                                                        | 1 | 0  | -           | 4         | -         | -     | -       | -   | -         | -         |
|            | SC1.110      | Science-1                                  | Full                | 3                                                        | 1 | 0  | -           | -         | -         | -     | 4       | -   | -         | -         |
|            | CS1.103      | Sports-3                                   | Full                | 0                                                        | 0 | 2  | 1           | -         | -         | -     | -       | -   | -         | -         |
|            |              | <b>Sub Total</b>                           |                     | 15                                                       | 5 | 5  | 1           | 13        | 0         | 4     | 4       | 0   | 0         | 0         |
|            | <b>Total</b> |                                            |                     |                                                          |   |    |             |           |           |       |         |     |           | <b>22</b> |

| Year & Sem | Course Code | Course Name                     | Full Sem / Half Sem | Lectures(L), Tutorial(T), Practical(P)<br>Hours per week |   |   | Credits (C) |           |           |       |         |     |           |      |
|------------|-------------|---------------------------------|---------------------|----------------------------------------------------------|---|---|-------------|-----------|-----------|-------|---------|-----|-----------|------|
|            |             |                                 |                     | L                                                        | T | P | Ins Core    | Prog Core | Prog Elec | Maths | Science | Hum | Open Elec | Hons |
|            |             |                                 |                     |                                                          |   |   |             |           |           |       |         |     |           |      |
| II-II      | Spring      |                                 |                     |                                                          |   |   |             |           |           |       |         |     |           |      |
|            | EC5.203     | Communication Theory            | Full                | 3                                                        | 1 | 0 | -           | 4         | -         | -     | -       | -   | -         | -    |
|            |             | Stream Foundation-1             | Full                | 3                                                        | 1 | 0 | -           | -         | 4         | -     | -       | -   | -         | -    |
|            | EC2.202     | Electronic Workshop-2           | Full                | 3                                                        | 1 | 3 | -           | 4         | -         | -     | -       | -   | -         | -    |
|            | HS8.102     | Intro to Human Sciences         | Full                | 3                                                        | 1 | 0 | -           | -         | -         | -     | -       | 4   | -         | -    |
|            | EC6.202     | Intro to Processer Architecture | Half                | 3                                                        | 1 | 0 | -           | 2         | -         | -     | -       | -   | -         | -    |
|            | OC1.104     | Sports-4                        | Full                | 0                                                        | 0 | 2 | 1           | -         | -         | -     | -       | -   | -         | -    |
|            | OC3.102     | Value Education-2               | Full                | 2                                                        | 1 | 0 | 2           | -         | -         | -     | -       | -   | -         | -    |
|            |             | Sub Total                       |                     | 17                                                       | 6 | 5 | 3           | 10        | 4         | 0     | 0       | 4   | 0         | 0    |
| Total      |             |                                 |                     |                                                          |   |   |             |           |           |       |         |     | 21        |      |
| III-I      | Monsoon     |                                 |                     |                                                          |   |   |             |           |           |       |         |     |           |      |
|            | EC9.302     | Stream Foundation-2             | Full                | 3                                                        | 1 | 0 | -           | -         | 4         | -     | -       | -   | -         | -    |
|            |             | Stream Elective-1               | Full                | 3                                                        | 1 | 0 | -           | -         | 4         | -     | -       | -   | -         | -    |
|            |             | Maths Elective                  | Full                | 3                                                        | 1 | 0 | -           | -         | -         | -     | -       | -   | 4         | -    |
|            |             | Open Elective-1                 | Full                | 3                                                        | 1 | 0 | -           | -         | -         | -     | -       | -   | 4         | -    |
|            |             | HSS Elective-1                  | Full                | 3                                                        | 1 | 0 | -           | -         | -         | -     | -       | 4   | -         | -    |
|            |             | Honours-1                       | Full                | 0                                                        | 2 | 6 | -           | -         | -         | -     | -       | -   | -         | 2    |
|            |             | Sub Total                       |                     | 15                                                       | 7 | 6 | 0           | 0         | 8         | 0     | 0       | 4   | 8         | 2    |
| Total      |             |                                 |                     |                                                          |   |   |             |           |           |       |         |     | 22        |      |
| III-II     | Spring      |                                 |                     |                                                          |   |   |             |           |           |       |         |     |           |      |
|            | SC1.111     | Open Elective-2                 | Full                | 3                                                        | 1 | 0 | -           | -         | -         | -     | -       | 4   | -         | -    |
|            |             | Science-2                       | Full                | 3                                                        | 1 | 0 | -           | -         | -         | -     | 4       | -   | -         | -    |
|            |             | Stream Elective-2               | Full                | 3                                                        | 1 | 0 | -           | -         | 4         | -     | -       | -   | -         | -    |
|            | EC9.303     | Honours-2                       | Full                | 0                                                        | 2 | 6 | -           | -         | -         | -     | -       | -   | -         | 2    |
|            |             | HSS Elective-2                  | Full                | 3                                                        | 1 | 0 | -           | -         | -         | 4     | -       | -   | -         | -    |
|            |             | Stream Elective-3               | Full                | 3                                                        | 1 | 0 | -           | -         | 4         | -     | -       | -   | -         | -    |
|            |             | Sub Total                       |                     | 15                                                       | 7 | 6 | 0           | 0         | 8         | 4     | 4       | 4   | 0         | 2    |
| Total      |             |                                 |                     |                                                          |   |   |             |           |           |       |         |     | 22        |      |



| Year &<br>Sem | Course<br>Code | Course Name | Full Sem<br>/ Half<br>Sem | Lectures(L),<br>Tutorial(T),<br>Practical(P)<br>Hours per week |   |   | Credits (C) |              |              |       |         |     |              |      |
|---------------|----------------|-------------|---------------------------|----------------------------------------------------------------|---|---|-------------|--------------|--------------|-------|---------|-----|--------------|------|
|               |                |             |                           | L                                                              | T | P | Ins<br>Core | Prog<br>Core | Prog<br>Elec | Maths | Science | Hum | Open<br>Elec | Hons |

|                     | Total      | %          |
|---------------------|------------|------------|
| Institute Core      | 12         | 7          |
| Programme Core      | 53         | 30         |
| Programme Electives | 32         | 18         |
| Maths               | 16         | 9          |
| Science             | 16         | 9          |
| Humanities          | 20         | 11         |
| Open Electives      | 20         | 11         |
| Honours             | 8          | 5          |
| <b>Total</b>        | <b>177</b> | <b>100</b> |